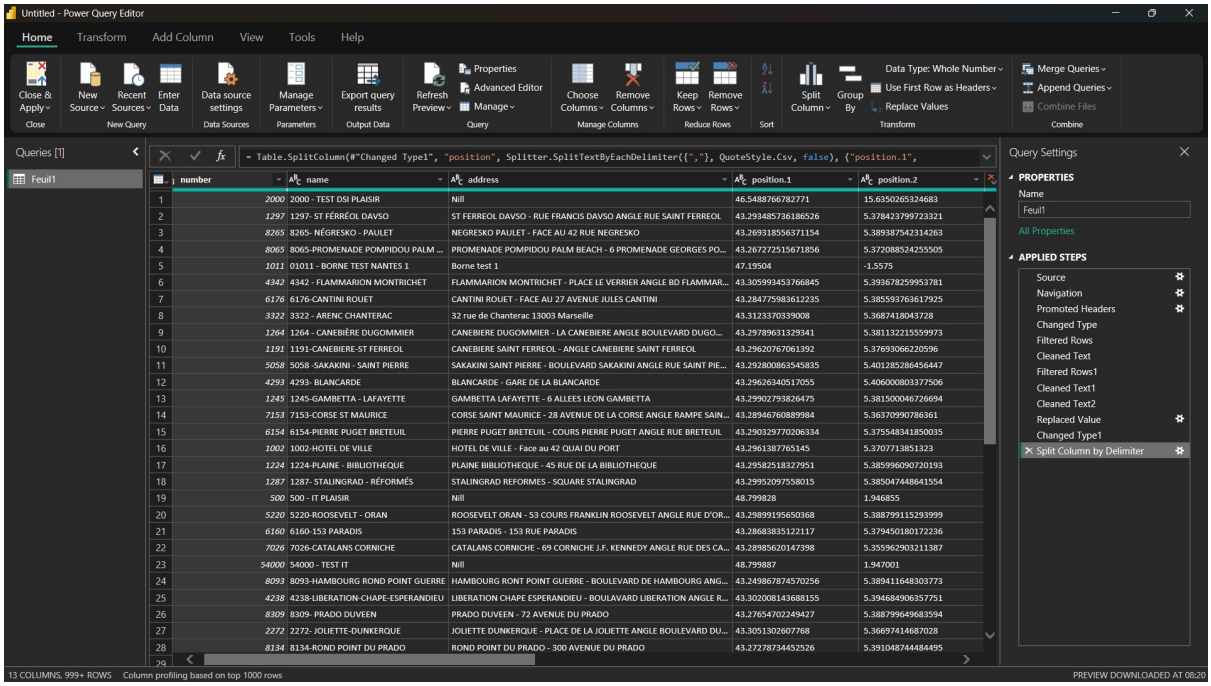
📚 Introduction

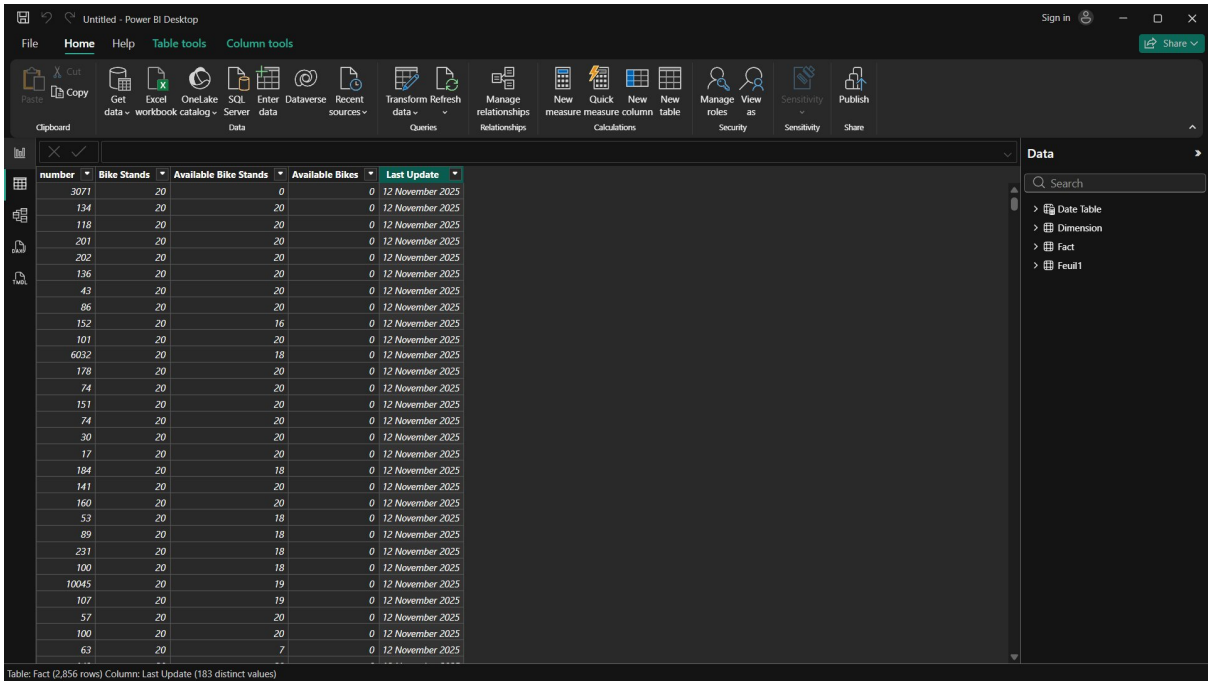
This dashboard provides a comprehensive overview of a **bike-sharing system's performance**, focusing on key operational and utilization metrics. It is designed to help stakeholders monitor how effectively the available infrastructure—such as bike capacity and docking stations—is being used.

The report highlights important indicators including **total capacity, available bikes, empty stands, occupancy rate, and availability percentage**, offering a clear picture of system efficiency. Additionally, it analyzes **station-wise distribution, contract-level performance, and year-wise trends**, enabling deeper insights into demand patterns and growth over time.

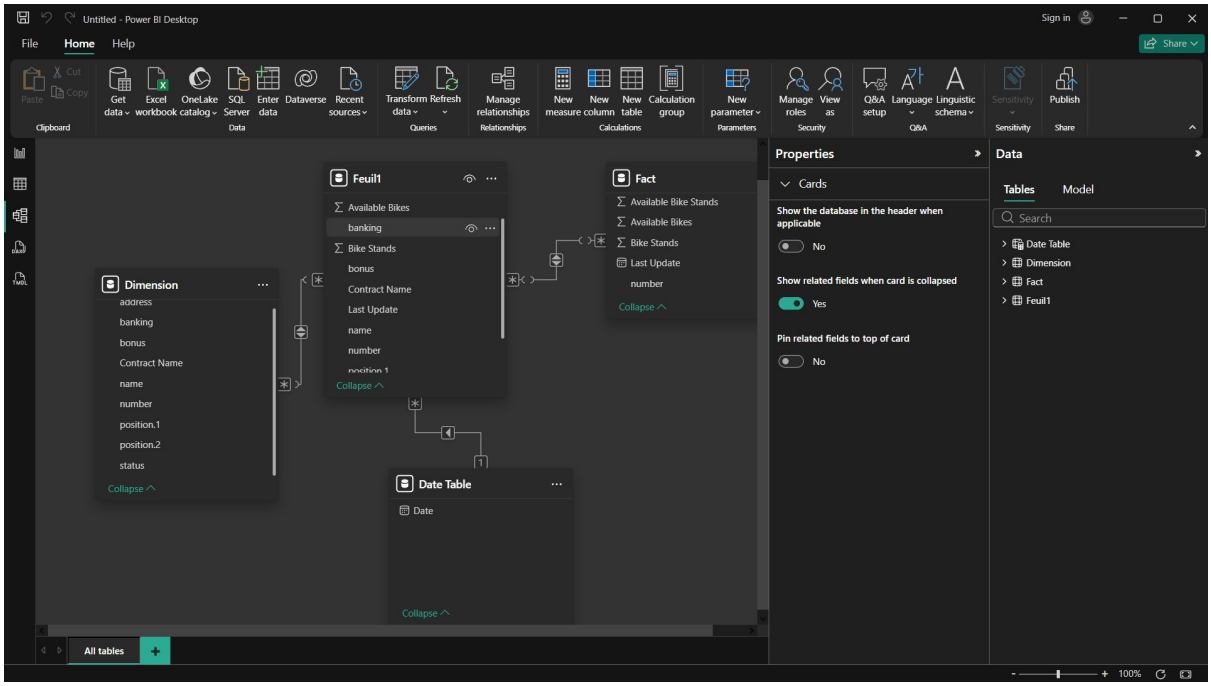
Overall, this dashboard serves as a decision-support tool to identify **underutilization, operational gaps, and opportunities for optimization**, helping improve service quality and maximize resource utilization.



Data cleaned through power query in power bi application.



Created Fact, Dimension and Date Table through Power Query



Relationship builds

📊 Dashboard Summary & Analysis (Bike Sharing System)

1. 🔢 Key KPIs Overview

- **Total Capacity:** 57K
- **Total Available Bikes:** 20K
- **Total Empty Stands:** 32K
- **Occupancy %: 34.34%**
- **Availability %: 56.15%**

👉 Insight:

- The system is **underutilized** (only ~34% occupancy).
- A large portion of infrastructure is still **empty stands**, indicating capacity is higher than current demand.

2. 🚲 Bike Availability Insights

- Available bikes are concentrated unevenly across stations.
- Highest station availability reaches around **54 bikes**, while most stations are around **35–40 bikes**.

👉 Insight:

- There is **no strong imbalance**, but a few stations have noticeably higher supply.
- Redistribution may improve efficiency.

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3. 🔎 Station-wise Performance

- Some stations (e.g., *INS*, *CAMINO*, *AVDA*, etc.) show higher availability compared to others.
- Lower-performing stations have slightly fewer bikes but still consistent range.

👉 Insight:

- System is **fairly balanced across stations**, but optimization is still possible.

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4. 📈 Trend Analysis (Year-wise)

- 2022–2023: Very low bike availability
- 2024: Slight increase
- 2025: Sharp spike to nearly **20K bikes**

👉 Insight:

- Strong **growth phase starting 2024–2025**
- Could indicate expansion, better data capture, or system scaling

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5. □ Occupancy vs Target

- **Current Occupancy:** 34.34%
- **Target:** 50%

👉 Gap:

- **~15.66% below target**

👉 Insight:

- The system is **not meeting utilization goals**
- Demand needs to increase or supply needs optimization

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6. 🗂 Contract-wise Distribution

- Major contribution from:
  - **Bruxelles-Capitale**
  - **Lyon**
  - **Toulouse**
- Smaller contributions spread across multiple cities (Dublin, Nantes, Sevilla, etc.)

👉 Insight:

- Usage is **highly concentrated in a few major regions**
- Smaller regions may need awareness or expansion strategies

7. 📊 Capacity Utilization

- Capacity (57K) is significantly higher than actual usage (20K bikes in use/available view)
- Large unused infrastructure gap

👉 Insight:

- Potential **over-investment in infrastructure**
- Opportunity to improve demand generation or redistribute assets

⚠ Key Problems Identified

- Low occupancy rate (34%)
- Uneven demand distribution across cities
- Large idle infrastructure (empty stands)
- Sudden spike in 2025 needs validation (data consistency check)

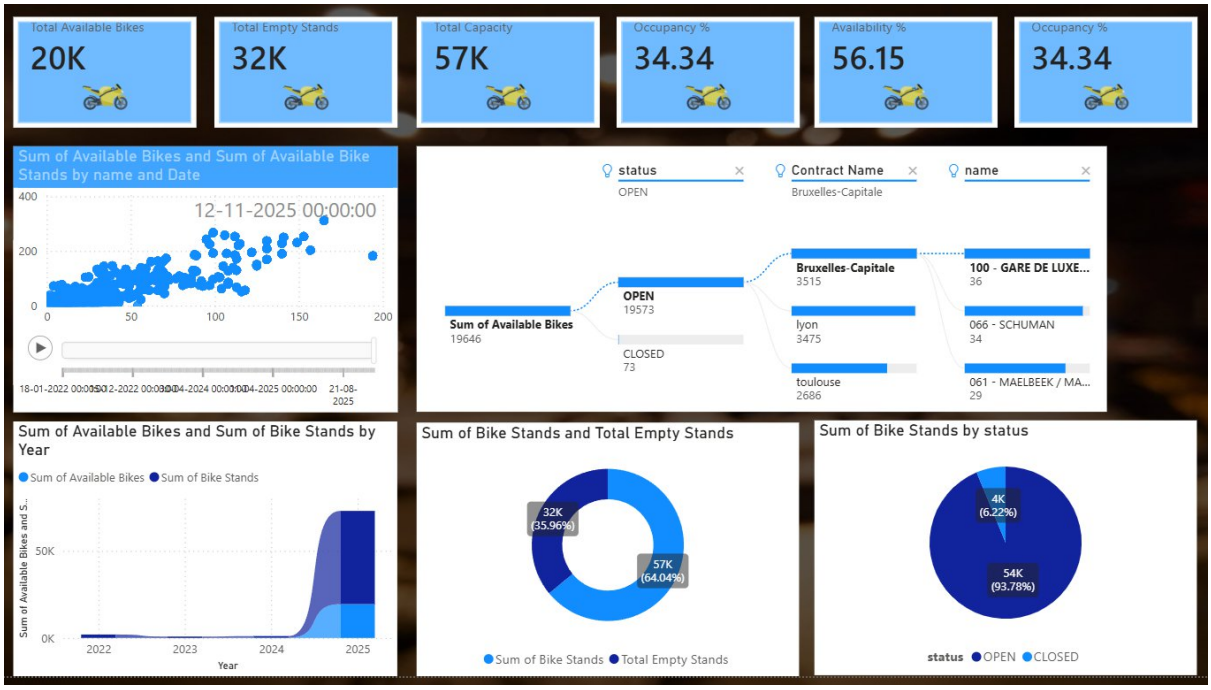
🚀 Recommendations

- Improve bike utilization through:
  - Dynamic pricing / incentives
  - Better station placement in low-demand areas

- Redistribute bikes from high-availability stations
- Investigate the **2025 spike** for data accuracy
- Focus marketing in low-performing regions

📌 Final Conclusion

The bike-sharing system has **strong infrastructure capacity but weak utilization**. While availability is good, occupancy is significantly below target, indicating the need for **demand improvement and operational optimization**.



📊 Short Summary

- The system has a **total capacity of 57K**, with **20K bikes available** and **32K empty stands**, indicating **low utilization**.
- **Occupancy is only 34.34%**, which is below the expected target, while **availability is relatively high (56.15%)**.
- Most stations are **operational (94% OPEN)**, with very few closed.
- A significant **increase in bikes and stands is seen in 2025**, showing rapid system expansion.
- Usage is **concentrated in major contracts like Bruxelles-Capitale, Lyon, and Toulouse**.